

No Benefit With Boosted Radiation Dose in Esophageal Cancer

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Increasing the dose of radiation therapy does not appear to improve outcomes for patients with esophageal cancer who are receiving definitive chemoradiation, according to new findings.

Overall, patients who received a standard dose of 50.4 Gy to the primary tumor and regional lymph nodes, combined with an integrated boost of 0.4 Gy per fraction (total 61.6 Gy), had no improvement in local control as compared with those receiving the standard dose alone.

The 3-year local progression-free survival (PFS) was 71% in the standard therapy arm vs 73% in the boosted arm, and 3-year overall survival was 41% vs 40%, respectively.

Lead author Maarten C.C.M. Hulshof, MD, PhD, of the Amsterdam University Medical Center, Netherlands, explained that in 2002, a previous dose escalation trial did not find an improvement in survival or local/regional control.

And now after "years of hard work, we are back to where we started, and the standard dose remains at 50 Gy," he said.

The study results were presented here at the 2020 Gastrointestinal Cancers Symposium.

During his presentation, Hulshof pointed out that the standard dose for local regional control in esophageal cancer is about 50 Gy, which is low compared to the dose for other solid tumor sites. As an example, it is 60 Gy to 80 Gy for malignancies of the larynx, lung, bladder, and cervix.

Similarly, local regional control rates are also lower as compared to other tumor sites (50% for the esophagus vs 70%-85% for larynx, bladder, and prostate).

Nearly 20 years ago, the INT 0123 (Radiation Therapy Oncology Group 94-05) dose escalation trial compared high-dose (64.8 Gy) with standard-dose (50.4 Gy) radiation therapy for the treatment of patients with esophageal cancer who were not undergoing surgery (*J Clin Oncol.* 2002;20[5]:1167-74). No significant difference was found in median survival, 2-year survival, or local/regional failure and local/regional persistence of disease.

"The trial was stopped prematurely, and negative results were never well understood," said Hulshof. "And so we thought there was a need for a dose-escalation trial, which we did in the

Netherlands."

The same biological dose was compared "with some innovations," he noted.

Study Details

In this study, 260 patients with clinical stage T2-4, N0-3, M0 carcinoma of the esophagus, and who were ineligible for surgery, were randomized to either a standard dose of 50.4 Gy or a high dose of 61.6 Gy.

The chemotherapy regimen was six concurrent cycles of carboplatin (AUC 2) and paclitaxel (50 mg/m²) for both arms.

The primary endpoint was to improve local PFS by 15%; secondary endpoints were locoregional PFS, overall survival, and toxicity. Patients were stratified for histological subtype, with squamous cell carcinoma comprising about 60% of tumors in both arms.